Braintap test (Physical Capability) - Insight 46 Phase 1

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Document details

Categories of variables	Physical Capability, Insight 46 Phase 1
Summary of work undertaken	The data from the website were reconciled to those transcribed manually by study staff. Impossible values have been coded as missing data. In several cases, the hand used for each test may have been recorded incorrectly (indicated by a discrepancy between the data recorded by staff and the website). This discrepancy has been coded with the variable “BT_rightleftuncertain_15“,the data is useable for analyses using a the mean of both hands, or “worst hand” analyses, but should be excluded if relationships with dominant or non-dominant hands are investigated. The website has provided some implausible values: Negative AT values: n=4 DS values of zero: n=23 The entire output for that hand has been discounted for these participants as no assurances could be given for the remainder of the statistics. **Missing data:** n= 16 did not complete the test, all due to equipment/ website malfunction n=1 only had valid data for one hand (so mean values missing) n=27 excluded due to implausible values (website error) **All re-coded as missing: 999** **Derived statistics:** Means for each hand
Output variables	bt_ds_lft15_i46p1, bt_at_lft15_i46p1, bt_ks_lft15_i46p1, bt_is_lft15_i46p1, bt_ds_rgt15_i46p1, bt_at_rgt15_i46p1, bt_ks_rgt_15_i46p1, bt_is_rgt_15_i46p1, bt_rgtlft15_i46p1, bt_at_mean15_i46p1, bt_ds_mean15_i46p1, bt_is_mean15_i46p1, bt_ks_mean15_i46p1, bt_website15_i46p1

Overview

Using a Lenovo Thinkpad laptop, participants were instructed to tap as fast and accurately as possible, alternating between the ‘s’ and ‘;’ keys for thirty seconds, with each hand separately. Data was analysed and stored on the website (, now defunct).

The following statistics were calculated automatically, for each hand separately: Kinesia Score (KS): the number of key taps in 30 seconds Akinesia Time (AT): mean dwell time on each key (milliseconds) Incoordination score (IS): variance of the interval between key taps (milliseconds)
Dysmetria Score (DS): ■a weighted index of the number of incorrect key taps, scored in a target fashion (1 point for the correct key, 2 points for immediately adjacent keys and 3 for other keys) then divided by the total number of key taps (i.e. if all keys are hit correctly, the score should be 1.0)

Data summary

Right Hand:

Variable	Obs	Mean	Std. Dev.	Min	Max

bt_at_right_15	464	90.64871	33.4036	32	262
bt_ds_right_15	464	1.05174	.0906383	1	2.0424
bt_is_right_15	464	39103.39	165800.3	274.5862	1955901
bt_ks_right_15	464	62.82112	11.7848	36	111

Left Hand:

Variable	Obs	Mean	Std. Dev.	Min	Max
bt_at_left_15	460	112.1457	39.19215	44	267
bt_ds_left_15	460	1.06227	.1111742	1	1.9675
bt_is_left_15	460	60436.6	423786.2	274.6021	7994885
bt_ks_left_15	460	58.45652	10.4471	24	94

Mean of R&L hands:

Variable	Obs	Mean	Std. Dev.	Min	Max
bt_at_mean_15	459	101.5163	33.63767	47	264.5
bt_ds_mean_15	459	1.055971	.08407	1	1.59495
bt_is_mean_15	459	49932.29	244949.5	299.4929	4160505
bt_ks_mean_15	459	60.66013	10.48054	30.5	94

Key publications

See (for example)

The Bradykinesia-Akinesia INoordination (BRAIN) Test is a validated, online test of motor speed. This was administered to Insight 46 participants during phase 1 (May 2015- Jan 2018).

The original description

Giovannoni, G., Van Schalkwyk, J., Fritz, V. U., & Lees, A. J. (1999). Bradykinesia akinesia inco-ordination test (BRAIN TEST): an objective computerised assessment of upper limb motor function. Journal of Neurology, Neurosurgery & Psychiatry, 67(5), 624-629.

A recent, larger validation

Noyce, A. J., Nagy, A., Acharya, S., Hadavi, S., Bestwick, J. P., Fearnley, J., ... & Giovannoni, G. (2014). Bradykinesia-akinesia incoordination test: validating an online keyboard test of upper limb function. PLoS One, 9(4).

Variables in this document

14 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Physical activity [6036] › Bradykinesia Akinesia Incoordination (BRAIN) Tap Test [60361]		
bt_at_lft15_i46p1	Akinesia time (AT) - Mean dwell time on each key (milliseconds) - left hand - Insight46 phase 1	topsheet field
bt_at_mean15_i46p1	Akinesia time (AT) - Mean dwell time on each key (milliseconds) - mean of left and right hands - Insight46 phase 1	topsheet field

Variable	Label	How it was found
bt_at_rgt15_i46p1	Akinesia time (AT) - Mean dwell time on each key (milliseconds) - right hand - Insight46 phase 1	topsheet field
bt_ds_lft15_i46p1	Dysmetria score (DS) - Weighted index of number of incorrect key taps, scored in a target fashion, then divided by the total number of key taps - left hand- Insight46 phase 1	topsheet field
bt_ds_mean15_i46p1	Dysmetria score (DS) - Weighted index of number of incorrect key taps, scored in a target fashion, then divided by the total number of key taps - mean of left and right hands - Insight46 phase 1	topsheet field
bt_ds_rgt15_i46p1	Dysmetria score (DS) - Weighted index of number of incorrect key taps, scored in a target fashion, then divided by the total number of key taps - right hand- Insight46 phase 1	topsheet field
bt_is_lft15_i46p1	Incoordination score (IS) - Variance of the interval between key taps (milliseconds) - left hand - Insight46 phase 1	topsheet field
bt_is_mean15_i46p1	Incoordination score (IS) - Variance of the interval between key taps (milliseconds) - mean of left and right hands - Insight46 phase 1	topsheet field
bt_is_rgt_15_i46p1	Incoordination score (IS) - Variance of the interval between key taps (milliseconds) - right hand - Insight46 phase 1	topsheet field
bt_ks_lft15_i46p1	Kinesia score (KS) - The number of taps in 30 seconds - left hand - Insight46 phase 1	topsheet field
bt_ks_mean15_i46p1	Kinesia score (KS) - The number of taps in 30 seconds - mean of left and right hands - Insight46 phase 1	topsheet field
bt_ks_rgt_15_i46p1	Kinesia score (KS) - The number of taps in 30 seconds - right hand - Insight46 phase 1	topsheet field
bt_rgtlft15_i46p1	Braintap - Hand labels possibly incorrect - Insight46 phase 1	topsheet field
bt_website15_i46p1	Brain tap - Implausible website values - Insight46 phase 1	topsheet field

2 low-confidence matches

Variable	Label	How it was found
Clinical data [51] › Physical measures [110] › Hand examination [1106]		
hand	Now I would like to examine your hands for any bumps or swellings. Would you be willing for me to examine your hands? - first record - at age 53 years	prose scan
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Heart and circulatory conditions [51014]		
limb	In the last ten years (that is since you were 43 years old,) have you had a sudden weakness in an arm or leg which got better after a day? - at age 53 years	prose scan